

Multisolving co-benefits for En-ROADS

Energy Supply

Coal

Taxing reduces air and water pollution, improving community and ecosystem health.



Oil

Taxing reduces chance of harmful oil spills. Improves national security & lowers military costs.



Natural Gas

Taxing improves water security & quality, protects wildlife and biodiversity.



Bioenergy

Taxing frees land for food production. Improves water & air quality, protects habitats.



Renewables

Subsidizing reduces air & water pollution, improves health, productivity, savings, energy access, & job opportunities.



Nuclear

Taxing reduces risk of exposure to radiation from nuclear meltdown or hazardous waste. Protects health of uranium miners.



New Zero Carbon

Research advancements in new technologies can create jobs and may be useful for other applications.



Carbon Price

Improves air quality, healthcare savings, & worker productivity. Makes renewable energy relatively cheaper. Funds can be earmarked for social programs.



Transport

Energy Efficiency

Increasing lowers energy costs. Improves public transit reduces traffic congestion & noise. Biking & walking increases physical activity & health savings.



Electrification

Increasing creates jobs in manufacturing & sales of electric batteries & engines. Improves air quality at the source, increasing health savings & worker productivity.



Buildings & Industry

Energy Efficiency

Increasing reduces energy demand & cost. Improves indoor air quality & health outcomes. Creates weatherization jobs.



Electrification

Increasing reduces noise pollution from motor engines, generators, & furnaces. Lowers energy costs. Improves indoor and outdoor air quality.



Growth

Population

Access to family planning, reproductive services, & education enhances quality of life for women.



Economic Growth

Low growth shifts focus from material consumption to alternative measures of wellbeing such as gross national happiness.



Carbon Dioxide Removal

Nature-based

Increasing creates jobs in tree planting & farming. Actions could improve soil health and ecosystems.



Technological

Carbon removal approaches can create jobs and support economic development in rural areas.



Other Sources of Greenhouse Gases

Agricultural Emissions

Reducing makes food production more efficient & less polluting, increasing food security & preserving ecosystem health.



Waste and Leakage

Reducing improves water and air pollution & protects the ozone layer. Fixing natural gas leaks can save money.



Deforestation

Reducing decreases erosion, & protects biodiversity & ecosystems. Preserves small-scale resource gathering & sustainable forestry livelihoods.



Equity Considerations for En-ROADS

Energy Supply

Coal

Taxing raises energy costs.  Low-income individuals rely on coal jobs yet suffer the most negative impacts of its production.

Oil

Industry protections must be eliminated.  Essential to provide low skill and high skill workers new job pathways.

Natural Gas

Poor communities & communities of color are disproportionately affected by impacts of drilling and burning. 

Bioenergy

Subsidizing may accelerate deforestation and can negatively impact farmer livelihoods by shifting agriculture markets. 

Renewables

Subsidy programs are usually limited to homeowners, leaving poorer communities unable to access the technology. 

Nuclear

Plants, mines, & waste sites often locate in low-income areas with limited resources for advocating stricter regulations. 

New Zero Carbon

There are unknown consequences and risks associated with new energy sources. 

Carbon Price

Fossil fuel workers risk losing their jobs. Higher costs may be passed on to consumer. Companies can find loop holes or exemptions due to corruption & rent-seeking. 

Transport

Energy Efficiency

High-quality pedestrian & cycling infrastructure is often concentrated in wealthier, white communities. Improved public transportation can improve social equity. 

Electrification

Electric vehicles and charging stations are not affordable or available to everyone. Lithium and copper mining severely harms ecosystems. 

Buildings & Industry

Energy Efficiency

High upfront costs of efficiency improvements. Policies often directed to property owners, excluding low-income renters from benefits. 

Electrification

High upfront costs of switching energy systems to electric. Household air pollution is unevenly distributed within and across countries. 

Growth

Population

Policies around limiting population growth should be voluntary, accessible, & empower women to make the choices that are best for them. 

Economic Growth

Gains in growth have gone to the world's wealthiest in recent decades. Policies must be tailored to specific local and regional circumstances. 

Carbon Dioxide Removal

Nature-based

Extensive land needs may compromise historical land access and compete with food production. Afforestation policies should avoid creating monocultures of trees. 

Technological

Many approaches have not yet been developed at scale and they pose unknown risks and consequences to nearby communities. 

Other Sources of Greenhouse Gases

Agricultural Emissions

Cultural values attached to certain foods. Local economies and employment that rely on agriculture can be threatened. 

Waste and Leakage

Some solutions add costs to goods, which may raise prices for consumers. F-gas alternatives may pose other safety concerns. 

Deforestation

Preservation efforts can limit indigenous access to lands they've sustainably inhabited for generations. Policies require engagement with local stakeholders. 